FEDERAL SPECIFICATION

# INSULATION BOARD, THERMAL, POLYURETHANE OR POLYISOCYANURATE FACED WITH ASPHALT/ORGANIC FELT, ASPHALT/ASBESTOS FELT OR ASPHALT/GLASS FIBER FELT ON BOTH SIDES OF THE FOAM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

This specification forms a part of the latest issue of Federal Specification HH-I-1972/GEN.

## 1 SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for faced thermal insulation boards composed of rigid cellular polyurethane or polyisocyanurate faced with asphalt/organic felts, asphalt/asbestos felts or asphalt/glass felts. This specification does not cover cryogenic applications. For intended use see 6.1 of HH-I-1972/GEN.

1.2 Classification. The insulation boards shall be of the following classes as specified (see 6.2 of HH-I-1972/GEN) and meet the physical requirements of 3.3, table I.

Class 1 - Compressive strength - 16 psi minimum.
Class 2 - Compressive strength - 35 psi minimum.
Class 3 - Compressive strength - 50 psi minimum.
Class 4 - Compressive strength - 75 psi minimum.

## 2. APPLICABLE DOCUMENTS

2.1 Latest issue of HH-I-1972/GEN and documents referenced therein.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 5640

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American Society for Testing and Materials (ASTM) Standards:

D226 - Asphalt-Saturated Organic Roofing Felt for Use in Membrane Waterproofing and Built-up Roofing.
D250 - Asphalt-Saturated Asbestos Roofing Felt for Use in Waterproofing and Constructing Built-up Roofs.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19102.)

3. REQUIREMENTS

3.1 Materials. The thermal insulation boards shall conform to the material requirements of the basic specification HH-I-1972/GEN and 3.1.1.

3.1.1 Facing materials. Unless otherwise specified (see 6.2 of HH-I-1972/GEN), the facing material incorporated in the design of the faced thermal insulation board shall be at the manufacturer's option one of the following, asphalt/organic, asphalt/asbestos, or asphalt/glass fiber felt on both sides of the foam. The facing material shall meet the requirements of 3.1.1.1 through 3.1.1.3.

3.1.1.1 Asphalt/organic felt The asphalt/organic felt shall conform to the material and physical properties requirements specified in table I of ASTM D226, except that the minimum saturation level shall be 95 percent.

3.1.1.2 Asphalt/asbestos felt. The asphalt/asbestos felt shall conform to the material and physical properties requirements specified in ASTM D250 except that the minimum saturation level shall be 15 percent.

3.1.1.3 Asphalt/glass fiber felt. The asphalt/glass fiber felt shall consist of fibrous glass felts saturated with asphalt or asphalt emulsion.

3.2 First article (see HH-I-1972/GEN).

3.3 Physical properties. The thermal insulation boards, when conditioned and tested in accordance with 4.5 through 4.5.7 of HH-I-1972/GEN shall meet the physical properties specified in table I.

TABLE I. Physical properties.

| Property | Class 1 | Class 2 | Class 3 | Class 4 | Test Reqmt. Para. |
|---|---|---|---|---|---|
| Compressive strength (psi) minimum | 16.0 | 35.0 | 50.0 | 75.0 | 4.5.3 |
| Dimensional stability (L) at -40° F/ambient RH,(W) (percent change) maximum | 2.0<br>2.0 | 2.0<br>2.0 | 2.0<br>2.0 | 2.0<br>2.0 | 4.5.4 |
| Dimensional stability (L) at 158° F/97% RH, (W) (percent change) maximum | 4.0<br>4.0 | 4.0<br>4.0 | 4.0<br>4.0 | 4.0<br>4.0 | 4.5.4 |
| Dimensional stability (L) at 200° F/ambient RH,(W) (percent change) maximum | 4.0<br>4.0 | 4.0<br>4.0 | 4.0<br>4.0 | 4.0<br>4.0 | 4.5.4 |



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4. QUALITY ASSURANCE PROVISIONS (see HH-I-1972/GEN)

5. PREPERATION FOR DELIVERY (see HH-I-1972/GEN).

6. NOTES

6.1 Intended use. The insulation material covered by this specification is intended for use as thermal insulation boards in roofs, and other applications where the insulation will not be subjected to temperatures below -40° F or above 200° F. The service temperature will vary depending on the application. This specification does not cover cryogenic applications. Consult manufacturer for recommendations and physical properties of faced polyurethane or polyisocyanurate insulation boards in cryogenic conditions.

6.2 Availability. Sizes - 3 feet by 4 feet, 4 feet by 4 feet by, 4 feet by 8 feet
Thicknesses - 3/4 to 4 inches.

CUSTODIANS:

Army - ME
Navy - YD
Air Force - 99

Review activity:

Army - CE

CIVIL AGENCY COORDINATING ACTIVITIES:

COM - NBS
GSA - FSS, PCD

Preparing activity:

Navy - YD

Project No. 5640-0281

U.S. GOVERNMENT PRINTING OFFICE : 1982 - 341-705/1268

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein.



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